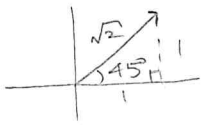
WORKSHEET 3.5: COPLANAR VECTORS

4a) $\vec{p} = -4\hat{i} + 5\hat{j}$

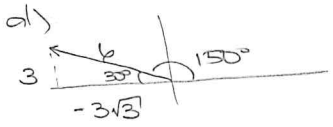
b) $\vec{OA} = 3\hat{i} - 3\hat{j}$

c)



$\vec{v} = \hat{i} + \hat{j}$

d)



$\vec{v} = -3\sqrt{3}\hat{i} + 3\hat{j}$

5a) $\vec{u} = (1,4)$, $\vec{v} = (-2,5)$ } not scalar multiples

$\therefore$ YES, they will form a basis in $\mathbb{R}^2$.

b) $(-567, -669) = a(1,4) + b(-2,5)$

① $-567 = a - 2b$ → ① $\times 4$ $\quad -2268 = 4a - 8b$

② $-669 = 4a + 5b$ $\quad$ ② $\quad -669 = 4a + 5b$

⊖ $\quad -1599 = -13b$

$123 = b$

sub $b = 123$ into ①.

$-567 = a - 2(123)$

$-321 = a$

$\therefore (-567, -669) = -321\vec{u} + 123\vec{v}$

6) $\vec{u} = (-4, -6) = -2(2,3)$

$\vec{v} = (10, 15) = 5(2,3)$

$\therefore \vec{u} = -\frac{2}{5}\vec{v}$

$\therefore$ scalar multiples $\therefore$ NO, they do not form a basis

7a) $\vec{u} = (1,-1,1)$
$\vec{v} = (0,1,1)$
$\vec{w} = (1,0,2)$

Assume $\vec{u} = a\vec{v} + b\vec{w}$

$(1,-1,1) = a(0,1,1) + b(1,0,2)$

① $1 = b$
② $-1 = a$
③ $1 = a + 2b$

} check in ③

LS = 1

RS = $a + 2b$
$= (-1) + 2(1)$
$= 1$

∴ LS = RS
∴ $\vec{u} = -\vec{v} + \vec{w}$
∴ the three vectors are coplanar

b) check triple scalar product

Let $\vec{u} = (1,0,1)$
$\vec{v} = (1,1,1)$
$\vec{w} = (1,0,-1)$

$\vec{v} \times \vec{w} = \begin{vmatrix} i & j & k \\ 1 & 1 & 1 \\ 1 & 0 & -1 \end{vmatrix}$

$= (-1, 2, -1)$

$\vec{u} \cdot (\vec{v} \times \vec{w})$
$= (1,0,1) \cdot (-1,2,-1)$
$= -1 + 0 - 1$
$= -2 \neq 0$

∴ they are not coplanar.

8a) i) $(4,2,2) = 2\vec{u}$

ii) $(1,2,4) = a(2,1,1) + b(-1,1,3)$

① $1 = 2a - b$
② $2 = a + b$
③ $4 = a + 3b$

} ① + ②

$3 = 3a$
$1 = a$

→ sub into ②
$2 = (1) + b$
$1 = b$

Check in ③

LS = 4

RS = $a + 3b$
$= (1) + 3(1)$
$= 4$

LS = RS ✓

∴ $(1,2,4) = \vec{u} + \vec{v}$

8a) iii) $(1, 5, 11) = a(2, 1, 1) + b(-1, 1, 3)$

① $1 = 2a - b$
② $5 = a + b$
③ $11 = a + 3b$

① + ② $\quad 6 = 3a$

$2 = a$

sub into ②

$5 = 2 + b$

$3 = b$

check in ③

LS = 11 $\quad$ RS = $2 + 3(3)$

$= 11$

LS = RS ✓

$\therefore (1, 5, 11) = 2\vec{u} + 3\vec{v}$

iv) $(4, 5, 8) = a(2, 1, 1) + b(-1, 1, 3)$

① $4 = 2a - b$
② $5 = a + b$
③ $8 = a + 3b$

① + ② $\quad 9 = 3a$

$3 = a$

sub into ②

$5 = 3 + b$

$2 = b$

Check in ③

LS = 8 $\quad$ RS = $3 + 3(2)$

$= 9$

LS ≠ RS

$\therefore$ Not possible.

b) Let $a = 1$, $b = 2$

$\vec{N} = 1(2, 1, 1) + 2(-1, 1, 3)$

$= (2, 1, 1) + (-2, 2, 6)$

$= (0, 3, 7)$

c) Let $a = 2$, $b = 2$.

$\vec{N} = 2(2, 1, 1) + 2(-1, 1, 3)$

$= (4, 2, 2) + (-2, 2, 6)$

$= (2, 4, 8)$

Change one component so that the a, b values won't work.

$\therefore \vec{N} = (2, 4, 9)$

9) $\vec{u} = x\vec{a} + 2y\vec{b}$

$\vec{v} = -2y\vec{a} + 3y\vec{b}$

$\vec{w} = 4\vec{a} - 2\vec{b}$

$\vec{w} = 2\vec{u} - \vec{v}$

$4\vec{a} - 2\vec{b} = 2(x\vec{a} + 2y\vec{b}) - (-2y\vec{a} + 3y\vec{b})$

$4\vec{a} - 2\vec{b} = 2x\vec{a} + 4y\vec{b} + 2y\vec{a} - 3y\vec{b}$

$4\vec{a} - 2\vec{b} = (2x + 2y)\vec{a} + (4y - 3y)\vec{b}$

$\therefore 4 = 2x + 2y$ & $-2 = y$

sub

$4 = 2x + 2(-2)$

$4 = 2x - 4$

$8 = 2x$

$4 = x$

$\therefore x = 4, y = -2.$

10a) $a(2,1,0) + b(-3,4,5) + c(2,0,3) = (-4,10,7)$

① $2a - 3b + 2c = -4$

② $a + 4b = 10$

③ $5b + 3c = 7.$

① $2a - 3b + 2c = -4$

② × 2 $2a + 8b = 20$

⊖ $-11b + 2c = -24.$ ④

③ × 2 $10b + 6c = 14$

④ × 3 $-33b + 6c = -72$

⊖ $43b = 86$

$b = 2$

Sub into ②

$a + 4(2) = 10$

$a = 2.$

Sub into ③

$5(2) + 3c = 7$

$3c = -3$

$c = -1$

$\therefore a = 2$

$b = 2$

$c = -1$

10b) $a(3,-1,2) + b(-1,1,3) + c(2,1,5) = (2,5,16)$

① $3a - b + 2c = 2$

② $-a + b + c = 5$

③ $2a + 3b + 5c = 16$

Elimnate b

① $3a - b + 2c = 2$

② $-a + b + c = 5$

⊕ $2a + 3c = 7$ ④

Eliminate b

② × 3 $-3a + 3b + 3c = 15$

③ $2a + 3b + 5c = 16$

⊖ $-5a - 2c = -1$ ⑤

④ × 2 $4a + 6c = 14$

⑤ × 3 $-15a - 6c = -3$

⊕ $-11a = 11$

$a = -1$

Sub $a = -1$ into ④

$2(-1) + 3c = 7$

$3c = 9$

$c = 3$

Sub $a = -1$, $c = 3$ into ①

$3a - b + 2c = 2$

$3(-1) - b + 2(3) = 2$

$1 = b$

$\therefore$ $a = -1$
$b = 1$
$c = 3$

11a) Let $\vec{u} = (1, -1, 1)$

$\vec{v} = (0, 1, 1)$

$\vec{w} = (1, 0, 2)$

$$\vec{v} \times \vec{w} = \begin{vmatrix} i & j & k \\ 0 & 1 & 1 \\ 1 & 0 & 2 \end{vmatrix}$$

$$= (2, 1, -1)$$

$$\vec{u} \cdot (\vec{v} \times \vec{w})$$

$$= (1, -1, 1) \cdot (2, 1, -1)$$

$$= 2 - 1 - 1$$

$$= 0$$

∴ Since $\vec{u} \cdot (\vec{v} \times \vec{w}) = 0$

the vectors are coplanar

b) Let $\vec{u} = (1, 0, 1)$

$\vec{v} = (1, 1, 1)$

$\vec{w} = (1, 0, -1)$

$$\vec{v} \times \vec{w} = \begin{vmatrix} i & j & k \\ 1 & 1 & 1 \\ 1 & 0 & -1 \end{vmatrix}$$

$$= (-1, 0, -1)$$

$$\vec{u} \cdot (\vec{v} \times \vec{w})$$

$$= (1, 0, 1) \cdot (-1, 0, -1)$$

$$= -1 + 0 - 1$$

$$= -2 \neq 0.$$

Since $\vec{u} \cdot (\vec{v} \times \vec{w}) \neq 0$,

the vectors are not coplanar